

# **CipherDocs: A Secure Solution to Prevent Forgery of Critical and Official Documents Using Polygon Blockchain-Based Encryption and Verification**

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## Introduction

The rapid increase in the use of digital documents across various domains such as education, governance, and legal systems has raised serious concerns regarding their security and authenticity. Traditional centralized systems are highly vulnerable to forgery, unauthorized modifications, and data breaches, as they rely on a single point of control. These systems often lack transparency and long-term trust, making it difficult to verify the integrity of documents. To address these challenges, blockchain technology offers a promising solution by providing immutability, transparency, and distributed trust. Therefore, there is a strong need for a secure and decentralized document verification mechanism that ensures authenticity, prevents tampering, and enhances overall reliability.



## Motivation

- High transaction costs in Ethereum-based solutions.
- Manual verification reduces efficiency and increases time.
- Static QR codes do not support real-time verification.
- Existing systems expose unnecessary sensitive data.
- Limited forgery detection capabilities.
- Need for automated, secure, and scalable system.

## Literature Survey

| <b>Sr. No.</b> | <b>Paper Name</b>                                                                            | <b>Core Technology</b> | <b>Specific Limitations Identified</b>                                                           | <b>CipherDocs Solution / Novelty</b>                                                                         |
|----------------|----------------------------------------------------------------------------------------------|------------------------|--------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| 1              | Certifier dapp-decentralized and secured certification system using blockchain               | Ethereum DApp          | High Gas Fees: Ethereum dependency makes it costly.<br>No AI: Purely hash-based verification.    | Polygon Network: Reduces fees significantly.<br>AI Integration: Adds semantic forgery detection.             |
| 2              | DOC-BLOCK: A blockchain based authentication system for digital documents                    | Ethereum + IPFS        | Binary Validation: No explanation for failure.<br>Privacy Leaks: Public IPFS content visibility. | Explainable AI: Provides reasons for rejection.<br>Context-Aware Encryption: Secures off-chain data.         |
| 3              | A Comparative Study of Ethereum and Polygon for Implementing NFT-Based Certification Systems | Eth vs. Polygon        | L2 Security Trade-offs: Notes security variance.<br>Complexity: Cross-chain management.          | Native Polygon Deployment: Optimizes specifically for L2 speed/cost, accepting trade-offs for usability.     |
| 4              | Docchain: Blockchain-based IoT solution for verification of degree documents.                | IoT + OCR              | OCR Fragility: Fails with poor scans/lighting.<br>Format Limits: B&W only.                       | Multimodal AI: Robust to image noise/artifacts.<br>Digital-First: Focuses on digital assets alongside scans. |



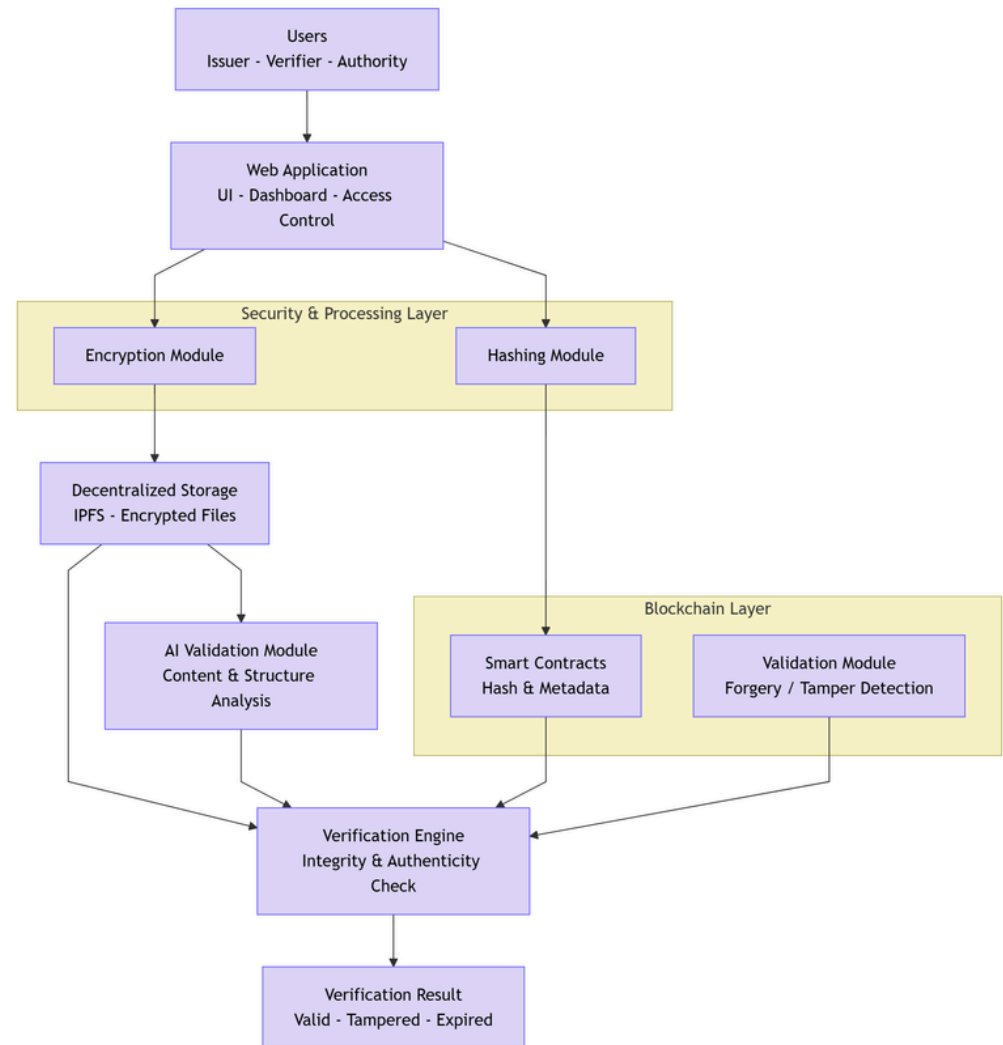
## Problem Statement & Objectives

Digital documents are highly vulnerable to forgery, unauthorized modifications, and lack of trust in centralized systems. Existing solutions are costly, inefficient, and do not provide real-time or secure verification mechanisms.

- Prevent document forgery and unauthorized alterations
- Reduce transaction costs in blockchain systems
- Enable real-time and automated document verification
- Ensure privacy-preserving validation mechanisms
- Support lifecycle management (issue, revoke, expire)
- Improve scalability and system efficiency

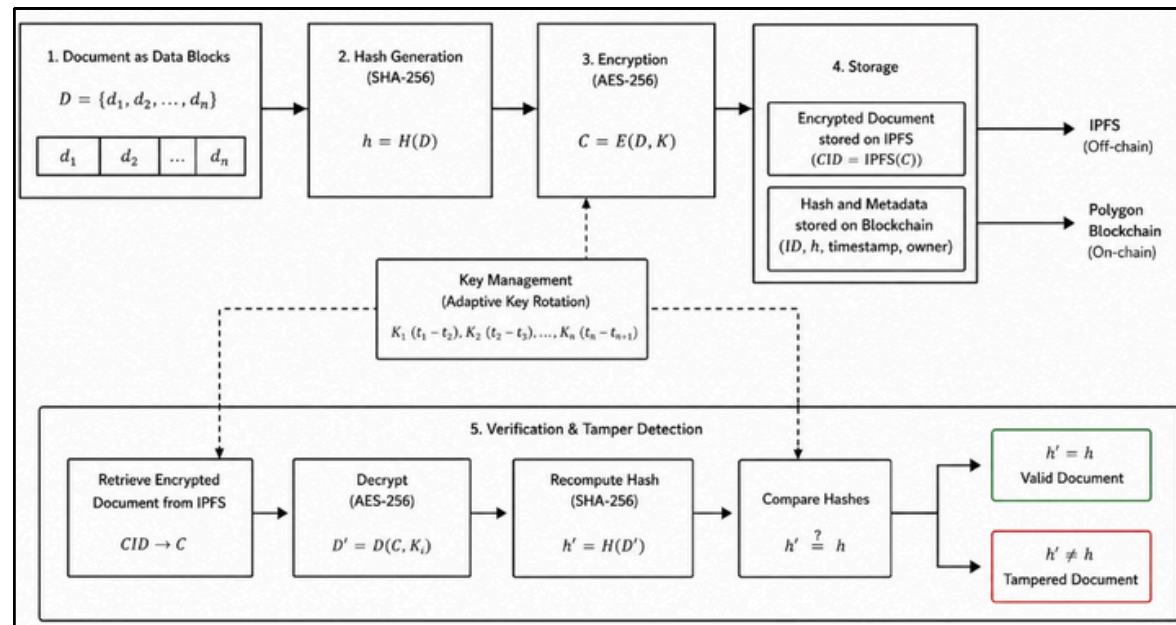
## Proposed System Architecture

- Decentralized architecture using Polygon blockchain and IPFS.
- AES-256 encryption applied before document storage.
- Encrypted documents stored securely on IPFS (off-chain).
- SHA-256 document hash stored on Polygon blockchain.
- Smart contracts manage issuance, verification, and revocation.
- Dynamic QR verification enables real-time authenticity checking.



## Cryptographic Model

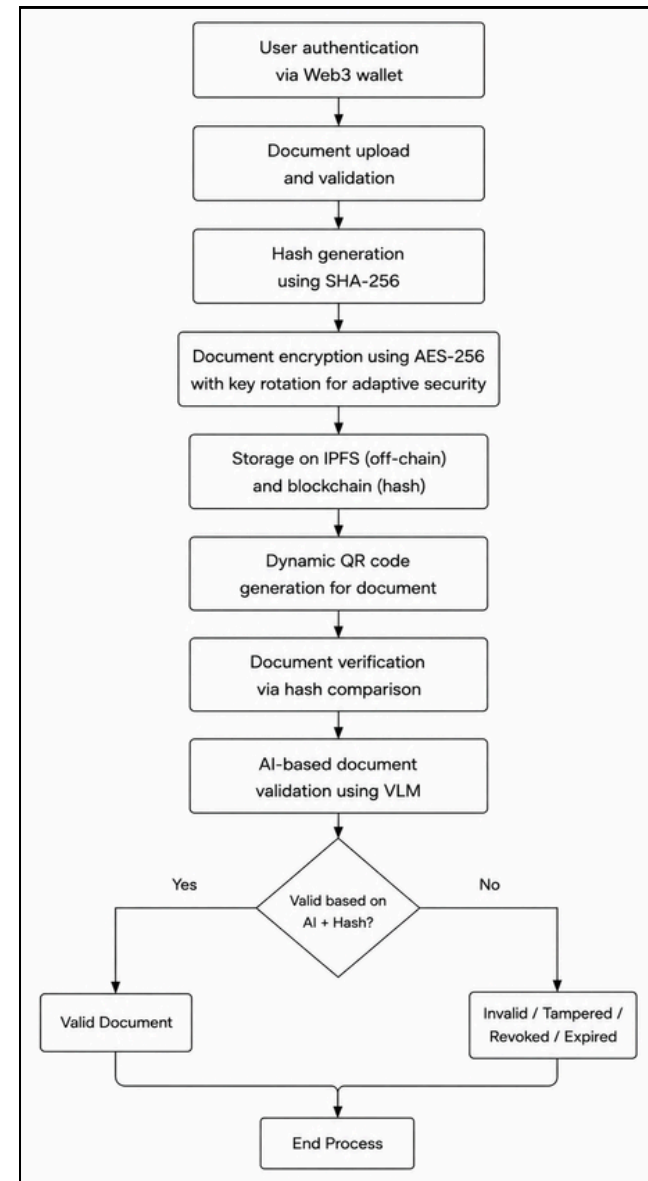
- Document represented as data blocks  $D = \{d_1, d_2, \dots, d_n\}$
- SHA-256 generates document hash  $h = H(D)$
- AES-256 encrypts document  $C = \text{AES}_{256}(D, K)$
- Adaptive key rotation improves encryption security
- Blockchain stores (ID, h, timestamp, owner)
- IPFS stores encrypted document securely
- Verification using recomputed hash  $h' = H(D')$
- $h' = h \rightarrow \text{Valid}$ , otherwise  $\rightarrow \text{Forged}$





## Algorithms / Workflow

- User authentication via Web3 wallet
- Document upload and validation
- SHA-256 hash generation
- AES-256 encryption with adaptive key rotation
- Encrypted document storage on IPFS (off-chain)
- Document hash storage on Polygon blockchain
- Dynamic QR code generation
- Document verification via hash comparison
- AI-based document validation using VLM
- Display result (Valid / Invalid / Revoked / Expired)



## Technology Stack

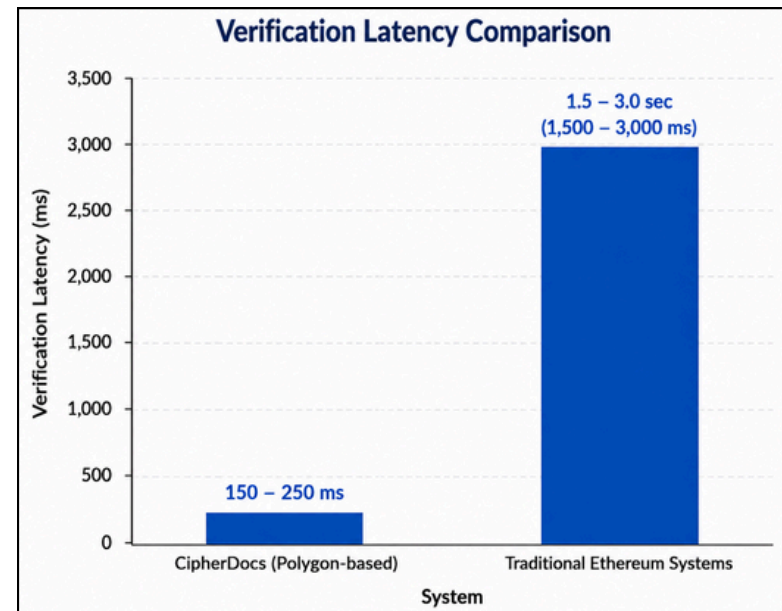
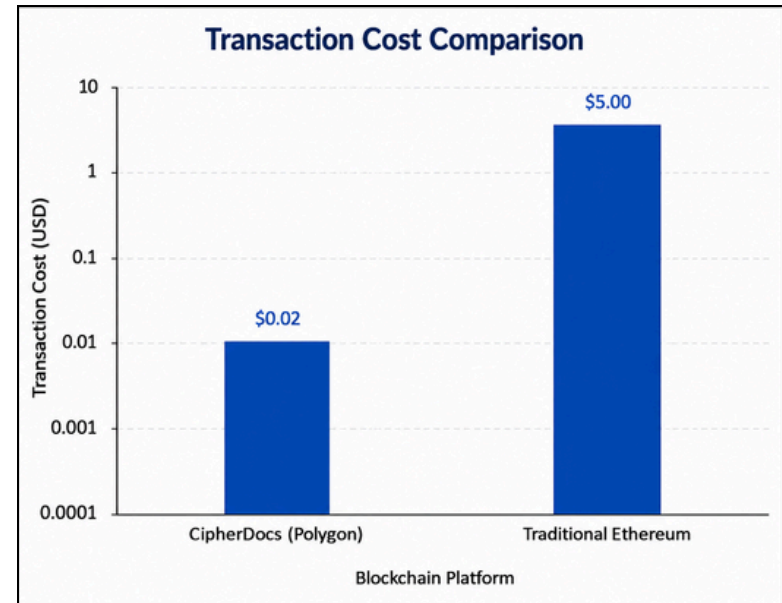
- Blockchain: Polygon (low cost, high scalability)
- Smart Contracts: Solidity
- Storage: IPFS (off-chain)
- Encryption: AES-256
- Hashing: SHA-256
- Authentication: Web3 wallets (MetaMask)
- AI validation module integrated
- Backend: Node.js, Express.js
- Frontend: HTML, CSS, JavaScript (Next.js)



# NEXT.js

## Performance Metrics

- Low Cost: ~\$0.001 – \$0.02 per transaction
- High Throughput: Scalable transaction processing
- Fast Verification: ~150 – 250 ms latency
- Scalable System: Supports large users and document volume
- High Security: SHA-256, AES-256, and tamper detection
- Efficient Storage: IPFS-based off-chain storage



## Results and Discussion

| Parameter            | CipherDocs (Polygon)                                       | Traditional Systems (Ethereum)   |
|----------------------|------------------------------------------------------------|----------------------------------|
| Transaction Cost     | \$0.0005 – \$0.02                                          | \$1 – \$50+                      |
| Throughput (TPS)     | ~120 – 540 TPS (practical)<br>Up to 2380 TPS (theoretical) | ~15 – 30 TPS                     |
| Verification Latency | 150 – 250 ms                                               | 1.5 – 3 sec                      |
| Block Time           | ~2 sec                                                     | ~12 sec                          |
| Scalability          | High                                                       | Moderate                         |
| Storage Method       | IPFS + Blockchain Hash                                     | Mostly On-chain / Centralized    |
| Security             | AES-256 + SHA-256 + Blockchain                             | Limited to Blockchain Only       |
| QR Verification      | Dynamic QR Verification                                    | Limited / Static QR Verification |



**Certificate is valid and authentic.** VERIFIED

ISSUER WALLET ADDRESS  
0xcae0a1144fe58397795e422786f076fa9c5a721c

USER WALLET ADDRESS  
0xf0771b82c234222fb00b5a86a5800348f114e880

CERTIFICATE ISSUED AT  
4/28/2026, 12:09:50 AM

BLOCKCHAIN TX  
0xb6b4c67c2cf691f093b02e630b0b1126a4dc082f6593d3dc13d2a8ad904b7b53

**Document has been tampered.** TAMPERED

Uploaded document does not match the originally issued certificate.

VERIFIED FILE  
sy-fake.pdf 210.4 KB

**Certificate has been expired.** EXPIRED

Certificate validity has expired.  
Expired On: 1/21/1970

VERIFIED FILE  
Rent Agreement.pdf 148.5 KB

**Certificate has been revoked.** REVOKED

Certificate has been revoked.  
Revoked At: 1/21/1970, 7:19:03 PM  
Expired On: 1/21/1970

VERIFIED FILE  
Land Record.pdf 2.3 KB

**AI Trust Score**  
Using the uploaded verification document to calculate trust score. Our AI will analyze semantic similarity, structure, and metadata to calculate a trust score (0-100).

**100**  
HIGH Trust Level  
✓ Document appears authentic

|                                                         |                                                  |                                            |
|---------------------------------------------------------|--------------------------------------------------|--------------------------------------------|
| <b>100</b><br>Text Similarity<br>Semantic content match | <b>100</b><br>Structure<br>Format & layout match | <b>100</b><br>Metadata<br>Properties match |
|---------------------------------------------------------|--------------------------------------------------|--------------------------------------------|

**Detailed Analysis**  
The document shows high authenticity.

MetaMask

Issuer

**Transaction request**

Network **A Amoy**

Request from cIPHERDOCS.VERCELAPP

Interacting with 0x9d368...Aa467

Network fee < \$0.01 P POL

Speed Site suggested ~2 sec

**AI-Powered Validation** TAMPERED

**Summary**  
The uploaded certificate has been marked as tampered by the cipherdocs verification system. This indicates that the certificate has been altered in some way, and its authenticity cannot be guaranteed.

**Verification Status**  
The verification status returned by cipherdocs is "tampered", indicating that the uploaded certificate does not match the original certificate stored in the system.

**Document Analysis**  
A comparison of the uploaded certificate text with the original certificate text stored in cipherdocs reveals several discrepancies. While the overall structure and content of the two certificates are similar, there are notable differences in certain fields.

**Detected Differences**  
The following differences were detected between the uploaded and original certificates:

- The ABC ID in the uploaded certificate is "949128680857", whereas in the original certificate, it is "948128680857".
- The course code and name for certain courses differ between the two certificates. For example, in Semester III, the course code "BIT3401" is listed as "DISCRETE MATHEMATICS" in the uploaded certificate, but as "FUNDAMENTALS OF DATA STRUCTURES" is present in the original, and "BIT3401" is actually "FUNDAMENTALS OF DATA STRUCTURES" in the original certificate, and "DISCRETE MATHEMATICS" is not found in the original.
- The grades and grade points for some courses also show discrepancies. However, without revealing full certificate content, it can be said that alterations are present in the uploaded document that are not reflected in the original.
- The cumulative CGPA in the uploaded certificate is "8.0", whereas in the original certificate, it is "7.6".

**AI-Powered Validation** VERIFIED

**Summary**  
The uploaded certificate has been verified against the original certificate stored in cipherdocs, and the result indicates that the certificate is valid.

**Verification Status**  
The status returned by cipherdocs is "valid", indicating that the uploaded certificate matches the original certificate stored in the system.

**Document Analysis**  
A thorough analysis of the uploaded certificate text and the original certificate text stored in cipherdocs has been conducted. The comparison reveals that the two documents are identical, with no discrepancies or modifications detected.

**Conclusion**  
Based on the analysis, it can be concluded that the uploaded certificate is authentic and has not been tampered with. The certificate matches the original certificate stored in cipherdocs, and all details, including student information, course grades, and semester records, are consistent. Therefore, the document trust status is confirmed to be valid.



## Conclusion

- CipherDocs provides a secure and decentralized document verification system using blockchain, IPFS, cryptographic techniques, and AI-based validation.
- Ensures document integrity, authenticity, and privacy while preventing unauthorized modifications and tampering.
- Enables automated and real-time verification with improved efficiency and reliability.
- Reduces operational cost through low-cost Polygon blockchain transactions.
- Offers a scalable solution suitable for education, governance, healthcare, and legal systems where secure document verification is critical.



## Future Work

- **Zero-Knowledge Proof (ZKP) Integration**

Enable privacy-preserving document verification without exposing sensitive document data.

- **Advanced AI-Based Forgery Detection**

Improve AI validation to detect sophisticated forgeries and AI-generated deepfake documents.

- **Multi-Blockchain Support**

Extend compatibility with blockchain platforms like Hyperledger and Solana for improved flexibility and interoperability.

- **Mobile Verification Support**

Enable mobile-based document verification and QR scanning for easier accessibility and real-time validation.





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